



## **BANNARI AMMAN INSTITUTE OF TECHNOLOGY**

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**22MA201**

### **ENGINEERING MATHEMATICS II**

#### **UNIT 2 & LP1 – SECOND ORDER LINEAR DIFFERENTIAL EQUATIONS**

##### **1. METHODS OF SOLVING SECOND ORDER LINEAR ORDINARY DIFFERENTIAL EQUATIONS:**

## Introduction

Linear differential equations are those in which the dependent variable and its derivatives occur only in the first degree and are not multiplied together. Thus the general Linear differential equation of the  $n^{\text{th}}$  order is of the form

$$\frac{d^n y}{dx^n} + p_1 \frac{d^{n-1} y}{dx^{n-1}} + p_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + p_n y = X$$

where  $p_1, p_2, \dots, p_n$  and  $X$  are functions of  $x$  only.

Linear differential equations with constant co-efficients are of the form

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + k_n y = X$$

where  $k_1, k_2, \dots, k_n$  are constants, such equations are most important in the study of electro-mechanical vibrations and other engineering problems.

## Example

1.  $4 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$
2.  $x^3 \frac{d^3 y}{dx^3} + 4x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 4y = x^2 + x$

## Order

The order of the highest differential coefficient that occurs in a differential equation is called the order of the differential equation.

## Degree

The degree of an equation is the power of the highest differential coefficient which occurs in it after the equation has been made rational and integral.

## Example

$$\left( \frac{d^2 y}{dx^2} \right)^3 + \left( \frac{d^3 y}{dx^3} \right)^2 + \frac{dy}{dx} - 4y = x^2 \sin x$$

Order is 3

Degree is 2

## Note

Denoting  $\frac{d}{dx}, \frac{d^2}{dx^2}, \frac{d^3}{dx^3}$  etc by  $D, D^2, D^3$  etc. So that  $\frac{dy}{dx} = D, \frac{d^2y}{dx^2} = D^2y, \frac{d^3y}{dx^3} = D^3y$

etc.

### Solution of Linear Differential Equation of $n^{\text{th}}$ order with constant coefficients

The given differential equation is

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = X$$

$$\text{(i.e.,)} \left[ a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n \right] y = X \quad (1)$$

The Solution of the differential equation is

$$\boxed{Y = CF + PI}$$

$Y =$  Complementary Function + Particular integral

#### Note

wherever  $X = 0$ , the Solution of the differential equation is  $Y = C \cdot F$  and whenever  $X = f(x)$ , the solution of the differential equation is

$$\boxed{Y = CF + PI}$$

### Rules to Finding the Complementary Function

#### Step 1

Form the auxiliary equation by replace  $D$  by  $m$  and  $x = 0$  in equation (1)

i.e., The auxiliary equation is

$$a_0 m^n + a_1 m^{n-1} + \dots + a_n = 0 \quad (2)$$

#### Step 2

Find the roots of the auxiliary equation.

Let  $m_1, m_2, \dots, m_n$  be the roots of equation (2)

#### Step 3

Case (i)

If all the roots be real and different, i.e.,  $m_1 \neq m_2 \neq \dots m_n$  then

$$C \cdot F = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$$

Case (ii)

If all the roots are equal (i.e.,)  $m = m_1 = m_2 = \dots = m_n$

$$C \cdot F = (c_1 x^{n-1} + c_2 x^{n-2} + \dots + c_n) e^{mx}$$

Case (iii)

If one pair of roots be imaginary, (i.e.,)  $m_1 = \alpha + i\beta$   $m_2 = \alpha - i\beta$  then

$$Y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x] + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

### Rules for Finding the Particular integral

Consider the equation (1) as  $\phi(D)y = x$

where

$$\phi(D) = a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n$$

$$P.I = \frac{1}{\phi(D)} \cdot X$$

$$P.I = \frac{1}{a_0 D^n + a_1 D^{n-1} + \dots + a_n} \cdot X$$

where  $X$  is a function of  $x$ .

Therefore the Solution of the given differential equation is

$$Y = CF + PI$$

### Note

1.  $\frac{1}{D} X = \int X dx$
2.  $\frac{1}{D-a} X = e^{ax} \int X e^{-ax} dx$

### Examples

1. Solve  $\frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 6x = 0$

#### Solution

Give  $\frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 6x = 0$

i.e.,  $(D^2 + 5D + 6)x = 0$

The auxiliary equation is

$$m^2 + 5m + 6 = 0$$

The roots are  $m = -3, -2$

$$(m+3)(m+2) = 0$$

∴ The solution is

$$Y = Ae^{-3x} + Be^{-2x}$$

2. Solve  $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$

Solution

The given equation is

$$[D^2 + 6D + 9]Y = 0$$

The auxiliary equation is

$$m^2 + 6m + 9 = 0$$

$$(m+3)(m+3) = 0$$

$$m = -3, -3$$

∴ The solution is

$$Y = [c_1x + c_2]e^{-3x}$$

3. Solve  $y'' - 2y' + 10y = 0$

Solution

The auxiliary equation is

$$m^2 - 2m + 10 = 0$$

$$m = \frac{2 \pm \sqrt{4 - 4(1)(10)}}{2} = \frac{2 \pm 2\sqrt{1-10}}{2}$$

$$m = 1 \pm 3i$$

$$\text{Here } \alpha = 1 \quad \beta = 3$$

The solution is  $y = e^x[A\cos 3x + B\sin 3x]$

4. Solve  $\frac{d^3y}{dx^3} + y = 0$

Solution

Given  $\frac{d^3y}{dx^3} + y = 0$

$$[D^3 + 1]y = 0$$

The auxiliary equation is

$$m^3 + 1 = 0 \Rightarrow (m+1)(m^2 - m + 1) = 0$$

$$m^3 = -1, \quad m = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm i\sqrt{3}}{2}$$

The roots are  $m = -1, \frac{1}{2} \pm \frac{i\sqrt{3}}{2}$

$\therefore$  The solution is

$$Y = Ae^{-x} + e^{\frac{x}{2}} \left[ A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right]$$

5. Solve  $[D^3 - 3D^2 + 3D - 1]y = 0$

Solution

Given  $[D^3 - 3D^2 + 3D - 1]y = 0$

The auxiliary equation is

$$m^3 - 3m^2 + 3m - 1 = 0$$

$$\begin{array}{cccc|c} 1 & -3 & 3 & -1 & \\ 0 & 1 & -2 & 1 & \\ \hline 1 & -2 & 1 & 0 & \end{array}$$

Put  $m = 1$

$$1 - 3 + 3 - 1 = 0$$

$\therefore 1$  is a root

$$m^2 - 2m + 1 = 0$$

$$(m-1)(m-1) = 0$$

The roots are  $m = 1, 1, 1$

The solution is

$$Y = [Ax^2 + Bx + C]e^x$$

### **Type I**

$$\phi(D)y = X$$

→ In this type  $X = e^{ax}$

$$PI = \frac{1}{\phi(D)} e^{ax} \quad a \text{ is a constant}$$

→ Replace 'D' 'a' ie  $D \rightarrow a$

$$PI = \frac{1}{\phi(a)} e^{ax}$$

→ If  $\phi(a) = 0$ , then

$$PI = x \cdot \frac{1}{\phi'(a)} e^{ax} \quad \text{provided } \phi'(a) \neq 0$$

→ If  $\phi'(a) = 0$  then

$$PI = x^2 \cdot \frac{1}{\phi''(a)} e^{ax} \quad \text{provided } \phi''(a) \neq 0$$

Continue the process till we get denominator of the RHS is not equal to 0.

### **Problems**

1. Solve  $(D^2 + 5D + 6)y = e^x$

#### **Solution**

The auxiliary equation is

$$m^2 + 5m + 6 = 0$$

$$(m+3)(m+2) = 0$$

$$m = -3, -2$$

$$\therefore CF = Ae^{-3x} + Be^{-2x}$$

$$PI = \frac{1}{D^2 + 5D + 6} e^x$$

$$= \frac{1}{1^2 + 5(1) + 6} e^x \quad \text{Replace } D \text{ by } 1$$

$$P \cdot I = \frac{e^x}{12}$$

The solution is  $Y = Ae^{-3x} + Be^{-2x} + \frac{e^x}{12}$

2. Solve  $[D^2 - 3D + 2]y = e^{3x}$

Solution

Given  $[D^2 - 3D + 2]y = e^{3x}$

The auxiliary equation is

$$m^2 - 3m + 2 = 0$$

$$(m - 2)(m - 1) = 0$$

$$m = 2, 1$$

$$\therefore \text{CF} = Ae^{2x} + Be^x$$

$$\begin{aligned} \text{PI} &= \frac{1}{D^2 - 3D + 2} e^{3x} \\ &= \frac{1}{(3)^2 - 3(3) + 2} e^{3x} \end{aligned}$$

Replace  $D \rightarrow 3$

$$= \frac{1}{9 - 9 + 2} e^{3x}$$

$$\text{PI} = \frac{e^{3x}}{2}$$

The Solution is  $Y = Ae^{2x} + Be^x + \frac{e^{3x}}{2}$

3. Solve  $(D^2 + 5D + 7)y = e^{\frac{-x}{2}} + 5$

Solution

Given  $(D^2 + 5D + 7)y = e^{\frac{-1}{2}x} + 5e^{0x}$

The auxiliary equation is

$$m^2 + 5m + 7 = 0$$

$$\begin{aligned} m &= \frac{-5 \pm \sqrt{25 - (4)(1)(7)}}{2} \\ &= \frac{-5 \pm \sqrt{25 - 28}}{2} = \frac{-5 \pm \sqrt{-3}}{2} \end{aligned}$$

$$m = \frac{-5 \pm 3i}{2} = \frac{-5}{2} \pm \frac{3}{2}i$$

$$\text{CF} = e^{\frac{-5}{2}x} \left[ A \cos \frac{3}{2}x + B \sin \frac{3}{2}x \right]$$



$$\begin{aligned}
 \text{PI}_1 &= \frac{1}{D^2 + 5D + 7} e^{\frac{-1}{2}x} \\
 &= \frac{1}{\left(\frac{-1}{2}\right)^2 + 5\left(\frac{-1}{2}\right) + 7} e^{\frac{-1}{2}x} \quad \text{Replace } D \rightarrow -\frac{1}{2} \\
 &= \frac{1}{\frac{1}{4} - \frac{5}{2} + 7} e^{\frac{-x}{2}} \\
 &= \frac{4}{1 - 10 + 28} e^{\frac{-x}{2}} = \frac{4}{19} e^{\frac{-x}{2}} \\
 \text{PI}_1 &= \frac{4}{19} e^{\frac{-x}{2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{PI}_2 &= \frac{1}{D^2 + 5D + 7} 5e^{0x} \\
 &= 5 \frac{1}{0 + 0 + 7} e^{0x} \quad \text{Replace } D \text{ by } 0 \\
 &= \frac{5}{7} e^{0x} \\
 \text{PI}_2 &= \frac{5}{7} \quad \because e^{0x} = 1
 \end{aligned}$$

The Solution is

$$Y = e^{\frac{-5}{2}x} \left[ A \cos \frac{3}{2}x + B \sin \frac{3}{2}x \right] + \frac{4}{19} e^{\frac{-x}{2}} + \frac{5}{7}$$

4. Solve  $[D^2 + D - 6]y = e^{3x} + e^{-3x}$

Solution

Given  $[D^2 + D - 6]y = e^{3x} + e^{-3x}$

The auxiliary equation is  $m^2 + m - 6 = 0$

$$(m + 3)(m - 2) = 0$$

$$m = -3, 2$$

$$\text{CF} = Ae^{-3x} + Be^{2x}$$

$$\begin{aligned}
 \text{PI}_1 &= \frac{1}{D^2 + D - 6} e^{3x} \\
 &= \frac{1}{9 + 3 - 6} e^{3x} \quad \text{Replace } D \rightarrow 3
 \end{aligned}$$

$$\text{PI}_1 = \frac{e^{3x}}{6}$$

$$PI_2 = \frac{1}{D^2 + D - 6} e^{-3x} = \frac{1}{9 - 3 - 6} e^{-3x}$$

Denominator is zero. Therefore we differentiate the Denominator with respect to D we get  
 $2D + 1$

$$PI_2 = x \cdot \frac{1}{2D + 1} e^{-3x}$$

$$= \frac{xe^{-3x}}{2(-3) + 1} \quad \text{Replace } D \rightarrow -3$$

$$PI_2 = \frac{xe^{-3x}}{-5}$$

The solution is  $y = Ae^{-3x} + Be^{2x} + \frac{e^{3x}}{6} - x \frac{e^{-3x}}{5}$

5. Solve  $[D^3 - 3D + 2]y = e^{-2x} + 2 \sinh x$

Solution

Given  $[D^3 - 3D + 2]y = e^{-2x} + 2 \sinh x$

The auxiliary equation is

$$m^3 - 3m + 2 = 0$$

Put  $m = 1$

$$1 - 3 + 2 = 0$$

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -3 & 2 \\ & 0 & 1 & +1 & -2 \\ \hline & 1 & 1 & -2 & \underline{0} \end{array}$$

$$0 = 0$$

$m = 1$  is a root

$$[m^2 + m - 2] = 0$$

$$(m + 2)(m - 1) = 0$$

The roots are  $-2, 1, 1$

$$CF = (Ax + B)e^x + Ce^{-2x}$$

$$\begin{aligned}
 \text{PI}_1 &= \frac{1}{D^3 - 3D + 2} e^{-2x} \\
 &= \frac{1}{-8 + 6 + 2} e^{-2x} \quad \text{Replace } D \rightarrow -2 \\
 &= x \frac{1}{3D^2 - 3} e^{-2x} \quad \because \text{Denominator is zero} \\
 &= x \frac{1}{3(4) - 3} e^{-2x} \quad \text{Replace } D \rightarrow -2 \\
 \text{PI}_1 &= \frac{xe^{-2x}}{9}
 \end{aligned}$$

$$\begin{aligned}
 \text{PI}_2 &= 2 \frac{1}{D^3 - 3D + 2} \sin hx \\
 &= 2 \frac{1}{D^3 - 3D + 2} \left[ \frac{e^x - e^{-x}}{2} \right] \quad \because \sinh x = \frac{e^x - e^{-x}}{2} \\
 &= \frac{1}{D^3 - 3D + 2} e^x - \frac{1}{D^3 - 3D + 2} e^{-x} \\
 &= \frac{e^x}{(1) - 3 + 2} - \frac{1}{(-1)^3 + 3 + 2} e^{-x} \\
 &= \frac{xe^x}{3D^2 - 3} - \frac{e^{-x}}{4} \\
 &= \frac{xe^x}{3(1) - 3} - \frac{e^{-x}}{4} \quad \text{Denominator is zero} \\
 &= \frac{x^2 e^x}{6D} - \frac{e^{-x}}{4} \quad \because D \rightarrow (1) \\
 \text{PI}_2 &= \frac{x^2 e^x}{6} - \frac{e^{-x}}{4}
 \end{aligned}$$

The solution is

$$Y = (Ax + B)e^x + Ce^{-2x} + \frac{xe^{-2x}}{9} + \frac{x^2 e^x}{6} - \frac{e^{-x}}{4}$$

6. Solve  $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 5y = -2 \cos hx$

Solution

$$\text{Given } [D^2 + 4D + 5]y = -2 \cos hx = -2 \left[ \frac{e^x + e^{-x}}{2} \right]$$

$$\because \cosh x = \frac{e^x + e^{-x}}{2}$$

The auxiliary equation is

$$m^2 + 4m + 5 = 0$$

$$m = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = \frac{-4 \pm 2i}{2}$$

$$m = -2 \pm i$$

$$CF = e^{-2x} [A \cos x + B \sin x]$$

$$\begin{aligned} PI &= \frac{1}{D^2 + 4D + 5} - [e^x + e^{-x}] \\ &= -\frac{1}{D^2 + 4D + 5} e^x - \frac{1}{D^2 + 4D + 5} e^{-x} \\ &= -\frac{1}{(1) + 4 + 5} e^x - \frac{1}{1 - 4 - 5} e^{-x} \quad \text{Replace } D \rightarrow +1 \text{ \& } D \rightarrow -1 \end{aligned}$$

$$PI_1 = -\frac{e^x}{10} - \frac{e^{-x}}{2}$$

The solution is

$$Y = e^{-2x} [A \cos x + B \sin x] - \frac{e^x}{10} - \frac{e^{-x}}{2}$$

$$7. \text{ Solve } (D^2 + D + 1)y = (1 + e^x)^2$$

Solution

$$\text{Given } (D^2 + D + 1)y = (1 + e^x)^2 = 1 + e^{2x} + 2e^x$$

The auxiliary equation is

$$m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm \sqrt{1 - 4(1)}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$$

$$m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$CF = e^{\frac{-x}{2}} \left[ A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right]$$

$$\begin{aligned}
 \text{PI} &= \frac{1}{D^2 + D + 1} (1 + e^{2x} + 2e^x) \\
 &= \frac{1}{D^2 + D + 1} e^{0x} + \frac{1}{D^2 + D + 1} e^{2x} + 2 \frac{1}{D^2 + D + 1} e^x \\
 &= \frac{e^{0x}}{1} + \frac{e^{2x}}{(2)^2 + 2 + 1} + \frac{2e^x}{(1) + (1) + 1} \\
 \text{PI} &= 1 + \frac{e^{2x}}{7} + \frac{2e^x}{3}
 \end{aligned}$$

The solution is

$$Y = e^{\frac{-x}{2}} \left( A \cos \frac{\sqrt{3}}{2} x + B \sin \frac{\sqrt{3}}{2} x \right) + 1 + \frac{e^{2x}}{7} + \frac{2e^x}{3}$$

8. Solve  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 10y = e^{5x+3}$

Solution

Given  $[D^2 - 3D - 10]y = e^{5x+3}$

The auxiliary equation is

$$m^2 - 3m - 10 = 0$$

$$(m - 5)(m + 2) = 0$$

$$m = 5, -2$$

$$\text{CF} = Ae^{5x} + Be^{-2x}$$

$$\begin{aligned}
 \text{PI} &= \frac{1}{D^2 - 3D - 10} e^{5x+3} \\
 &= \frac{1}{25 - 15 - 10} e^{5x+3} \quad [\text{Replace } D \text{ by } 5 \because \text{coefficient of } x = 5] \\
 &= \frac{x}{2D - 3} e^{5x+3} \quad [\because \text{Denominator is zero in previous step}] \\
 \text{PI} &= \frac{xe^{5x+3}}{10 - 3} = \frac{xe^{5x+3}}{7}
 \end{aligned}$$

The Solution is  $Y = Ae^{5x} + Be^{-2x} + \frac{x}{7} e^{5x+3}$

9. Solve  $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

Solution

Given  $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

The auxiliary equation is

$$m^3 + 3m^2 + 3m + 1 = 0$$

Put  $m = -1$

$$-1 + 3 - 3 + 1 = 0 \quad \begin{array}{r|rrrr} -1 & 1 & 3 & 3 & 1 \\ & 0 & -1 & -2 & -1 \\ \hline & 1 & 2 & 1 & \underline{0} \end{array}$$

$$0 = 0$$

$m = -1$  is a root

$$m^2 + 2m + 1 = 0$$

$$(m+1)(m+1) = 0$$

The roots are  $m = -1, -1, -1$

$$CF = (Ax^2 + Bx + c)e^{-x}$$

$$\begin{aligned} PI &= \frac{1}{D^3 + 3D + 3D + 1} e^{-x} \\ &= \frac{e^{-x}}{-1 + 3 - 3 + 1} \quad \begin{array}{l} \text{Replace } D \rightarrow -1 \\ \text{Denominator is zero} \end{array} \\ &= \frac{xe^{-x}}{3D^2 + 6D + 3} \\ &= \frac{xe^{-x}}{3 - 6 + 3} \quad \text{Denominator is zero} \\ &= \frac{x^2 e^{-x}}{6D + 6} = \frac{x^2 e^{-x}}{-6 + 6} \quad \text{Denominator is zero} \\ PI &= \frac{x^3 e^{-x}}{6} \end{aligned}$$

The Solution is  $Y = (Ax^2 + Bx + C)e^{-x} + \frac{x^3 e^{-x}}{6}$

10. Solve  $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$

Solution

Given  $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$

The auxiliary equation is

$$m^3 - 2m^2 - 5m + 6 = 0 \quad \begin{array}{r|rrrr} 1 & 1 & -2 & -5 & 6 \\ & 0 & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & \underline{0} \end{array}$$

Put  $m = 1$   $1 - 2 - 5 + 6 = 0$

$m = 1$  is a root

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

The roots are  $m = 1, 3, -2$

$$CF = Ae^x + Be^{3x} + Ce^{-2x}$$

$$PI = \frac{1}{D^3 - 2D^2 - 5D + 6} e^{3x}$$

$$= \frac{1}{27 - 18 - 15 + 6} e^{3x} \quad \text{Denominator is zero}$$

$$= \frac{x}{3D^2 - 4D - 5} e^{3x}$$

$$= \frac{xe^{3x}}{27 - 12 - 5}$$

$$PI = \frac{xe^{3x}}{10}$$

The Solution is

$$Y = Ae^x + Be^{3x} + Ce^{-2x} + \frac{xe^{3x}}{10}$$

## **Type – II**

$\phi(D)y = X$  Here  $X = \sin(ax + b)$  or  $\cos(ax + b)$

$$\rightarrow PI = \frac{1}{\phi(D)y} [\sin(ax + b)] \text{ or } [\cos(ax + b)]$$

$\rightarrow$  Replace  $D^2$  by  $-a^2$

$\rightarrow$  If denominator reduces to a constant it is the P.I

$\rightarrow$  If denominator reduces into  $D$  only, the operate it into the given function  $\sin(ax + b)$  or  $\cos(ax + b)$

$$\therefore \frac{1}{D} = \int dx$$

$\rightarrow$  If denominator reduces to a factor of the form  $\alpha D + \beta$  then we apply conjugate multiplication  $\alpha D - \beta$  on both numerator and denominator from left hand side. Then replace  $D^2$  by  $-a^2$  in the denominator to get a constant.

$\rightarrow$  If the denominator is zero then differentiate the denominator with respect to  $D$  and multiply it by  $x$  and continue the replacement work

This process will be repeated till we reach a constant in the denominator or any value other than zero.

### **Problems**

1. Solve  $(D^2 - 4D + 4)y = 8\sin 2x$

#### **Solution**

Given  $(D^2 - 4D + 4)y = 8\sin 2x$

The Auxiliary equation is

$$m^2 - 4m + 4 = 0$$

$$(m - 2)(m - 2) = 0$$

The roots are  $m = 2, 2$

$$CF = (Ax + B)e^{2x}$$

$$\begin{aligned} PI &= \frac{1}{D^2 - 4D + 4} 8\sin 2x \\ &= \frac{1}{-4 - 4D + 4} 8\sin 2x \quad \text{Replace } D^2 \rightarrow -(2)^2 \\ &= \frac{1}{-4D} 8\sin 2x \\ &= -\frac{1}{D} 2\sin 2x \\ &= -2 \frac{(-\cos 2x)}{2} \quad \because \frac{1}{D} \sin 2x = \int \sin 2x \, dx = \frac{-\cos 2x}{2} \end{aligned}$$

$$PI = \cos 2x$$

The Solution is

$$Y = (Ax + B)e^{2x} + \cos 2x$$

2. Solve  $(D^2 + 1)y = \sin x \sin 2x$

#### **Solution**

Given  $(D^2 + 1)y = \sin x \sin 2x$

The auxiliary equation is  $m^2 + 1 = 0$

$$m^2 = -1 \quad m = \pm i$$

$$CF = A \cos x + B \sin x$$



$$\begin{aligned}
PI &= \frac{1}{D^2+1} \sin x \sin 2x \quad \sin A \sin B = \frac{\cos(A-B) - \cos(A+B)}{2} \\
&= \frac{1}{D^2+1} \left[ \frac{\cos x - \cos 3x}{2} \right] \\
&= \frac{1}{2} \left[ \frac{1}{D^2+1} \cos x - \frac{1}{D^2+1} \cos 3x \right] \\
&= \frac{1}{2} \left[ \frac{1}{(-1)+1} \cos x - \frac{1}{-9+1} \cos 3x \right] \\
&= \frac{1}{2} \left[ x \frac{1}{2D} \cos x + \frac{1}{8} \cos 3x \right] \\
&= \frac{1}{2} \left[ \frac{x}{2} \sin x + \frac{1}{8} \cos 3x \right] \\
PI &= \frac{x \sin x}{4} + \frac{\cos 3x}{16}
\end{aligned}$$

The Solution is

$$Y = A \cos x + B \sin x + \frac{x \sin x}{4} + \frac{\cos 3x}{16}$$

3. Solve  $(D^3 - 1)y = \cos(2x - 1)$

Solution

Given  $(D^3 - 1)y = \cos(2x - 1)$

A. E is  $m^3 - 1^3 = 0$

$$(m-1)(m^2+m+1)=0$$

$$m=1, m^2+m+1=0$$

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

The roots are  $m = 1, \frac{-1 \pm i\sqrt{3}}{2}$

$$C.F = Ae^x + e^{\frac{-x}{2}} \left[ B \cos \frac{\sqrt{3}}{2} x + C \sin \frac{\sqrt{3}}{2} x \right]$$

$$\begin{aligned}
PI &= \frac{1}{D^3 - 1} \cos(2x - 1) \\
&= \frac{1}{D(D^2) - 1} \cos(2x - 1) \\
&= \frac{1}{-4D - 1} \cos(2x - 1) \quad \text{Replace } D^2 \rightarrow -(2)^2 \\
&= -\frac{1}{4D - 1} \cos(2x - 1) \\
&= -\frac{4D + 1}{(4D - 1)(4D + 1)} \cos(2x - 1) \\
&= -\frac{(4D + 1)}{16D^2 - 1} \cos(2x - 1) \\
&= -\frac{(4D + 1)}{16(-4) - 1} \cos(2x - 1) \quad \text{Replace } D^2 \rightarrow -(2)^2 \\
&= -\frac{1}{-65} [4D(\cos(2x - 1)) + \cos(2x - 1)] \\
&= \frac{1}{65} [4(-2 \sin(2x - 1)) + \cos(2x - 1)] \\
PI &= \frac{1}{65} [-8 \sin(2x - 1) + \cos(2x - 1)]
\end{aligned}$$

The Solution is

$$Y = Ae^x + e^{\frac{-x}{2}} \left[ B \cos \frac{\sqrt{3}}{2} x + C \sin \frac{\sqrt{3}}{2} x \right]$$

4. Solve  $(D^2 + 16)y = \cos^3 x$

Solution

Given  $(D^2 + 16)y = \cos^3 x$

A. E is  $m^2 + 16 = 0$

$$m^2 = -16 \quad m = \pm 4i$$

$$CF = A \cos 4x + B \sin 4x$$

$$\begin{aligned}
PI &= \frac{1}{D^2 + 16} \cos^3 x \\
&= \frac{1}{D^2 + 16} \left[ \frac{\cos 3x + 3 \cos x}{4} \right] \because \cos^3 x = \frac{\cos 3x + 3 \cos x}{4} \\
&= \frac{1}{4} \left[ \frac{1}{D^2 + 16} \cos 3x + \frac{1}{D^2 + 16} 3 \cos x \right] \\
&= \frac{1}{4} \left[ \frac{1}{-9 + 16} \cos 3x + 3 \frac{1}{-1 + 16} \cos x \right] \text{ Replace } D^2 \rightarrow (-3^2) \text{ or } D^2 \rightarrow (-1^2) \\
&= \frac{1}{4} \left[ \frac{1}{7} \cos 3x + \frac{1}{5} \cos x \right] \\
PI &= \frac{1}{28} \cos 3x + \frac{1}{20} \cos x
\end{aligned}$$

$$\text{The Solution } Y = A \cos 4x + B \sin 4x + \frac{\cos 3x}{28} + \frac{\cos x}{20}$$

$$5. \text{ Solve } (3D^2 + D - 14)y = \cos x$$

Solution

$$\text{Given } (3D^2 + D - 14)y = \cos x$$

$$\text{A.E is } 3m^2 + m - 14 = 0$$

$$(m - 2) \left( m + \frac{7}{3} \right) = 0$$

The roots are 2,  $-\frac{7}{3}$

$$CF = Ae^{2x} + Be^{-\frac{7}{3}x}$$

$$\begin{aligned}
PI &= \frac{1}{3D^2 + D - 14} \cos x \\
&= \frac{1}{3(-1) + D - 14} \cos x \quad \text{Replace } D^2 \rightarrow -(1)^2 \\
&= \frac{1}{D - 17} \cos x \\
&= \frac{D + 17}{(D - 17)(D + 17)} \cos x = \frac{D + 17}{D^2 - (17)^2} \cos x \\
&= \frac{D(\cos x) + 17 \cos x}{(-1) - 289}
\end{aligned}$$

$$PI = -\frac{\sin x + 17 \cos x}{290}$$

The Solution

$$Y = Ae^{2x} + Be^{\frac{-7x}{3}} - \frac{\sin x + 17 \cos x}{290}$$

6. Solve  $(D^2 - 4)y = \sin^2 x$

Solution

Given  $(D^2 - 4)y = \sin^2 x$

A.E is  $m^2 - 4 = 0$

$$m^2 = 4 \quad m = \pm 2$$

$$CF = Ae^{2x} + Be^{-2x}$$

$$\begin{aligned} PI &= \frac{1}{D^2 - 4} \left[ \frac{1 - \cos 2x}{2} \right] \because \sin^2 x = \frac{1 - \cos 2x}{2} \\ &= \frac{1}{2} \left[ \frac{1}{D^2 - 4} [1 - \cos 2x] \right] \\ &= \frac{1}{2} \left[ \frac{e^{0x}}{D^2 - 4} - \frac{1}{D^2 - 4} \cos 2x \right] \because e^{0x} = 1 \\ &= \frac{1}{2} \left[ \frac{1}{-4} e^{0x} - \frac{1}{-4 - 4} \cos 2x \right] \\ &= \frac{1}{2} \left[ \frac{1}{-4} e^{0x} - \frac{1}{-4 - 4} \cos 2x \right] \end{aligned}$$

$$PI = -\frac{1}{8} + \frac{1}{16} \cos 2x$$

The Solution is

$$Y = Ae^{2x} + Be^{-2x} - \frac{1}{8} + \frac{1}{16} \cos 2x$$

7. Solve  $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

Solution

Given  $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

A.E is  $m^2 - 4m + 3 = 0$

$$(m - 3)(m - 1) = 0$$

The roots are  $m = 1, 3$

$$CF = Ae^x + Be^{3x}$$

$$\begin{aligned}
PI &= \frac{1}{D^2 - 4D + 3} \sin 3x \cos 2x \left[ \because \sin A \cos B = \frac{\sin(A+B) + \sin(A-B)}{2} \right] \\
&= \frac{1}{D^2 - 4D + 3} \left[ \frac{\sin 5x + \sin x}{2} \right] \\
&= \frac{1}{2} \left[ \frac{1}{D^2 - 4D + 3} \sin 5x + \frac{1}{D^2 - 4D + 3} \sin x \right] \\
&= \frac{1}{2} \left[ \frac{1}{-25 - 4D + 3} \sin 5x + \frac{1}{-1 - 4D + 3} \sin x \right] \quad \begin{array}{l} \text{Replace } D^2 \rightarrow -5^2 \\ D^2 \rightarrow -1^2 \end{array} \\
&= \frac{1}{2} \left[ \frac{1}{-4D - 22} \sin 5x + \frac{1}{-4D + 2} \sin x \right] \\
&= \frac{1}{2} \left[ \frac{-4D + 22}{(-4D - 22)(-4D + 22)} \sin 5x + \frac{-4D - 2}{(-4D + 2)(-4D - 2)} \sin x \right] \\
&= \frac{1}{2} \left[ \frac{-4D(\sin 5x) + 22 \sin 5x}{16D^2 - (22)^2} + \frac{[-4D(\sin x) - 2 \sin x]}{16D^2 - 4} \right] \\
&= \frac{1}{2} \left[ \frac{-4(5 \cos 5x) + 22 \sin 5x}{16(-25) - 484} - \frac{[4(\cos x) + 2 \sin x]}{-20} \right] \\
&= \frac{1}{2} \left[ \frac{-20 \cos 5x + 22 \sin 5x}{-400 - 484} + \frac{4 \cos x + 2 \sin x}{20} \right] \\
&= \frac{1}{2} \left[ \frac{-20 \cos 5x + 22 \sin 5x}{-884} + \frac{2 \cos x + \sin x}{10} \right] \\
PI &= \frac{1}{2} \left[ \frac{-10 \cos 5x + 11 \sin 5x}{-442} + \frac{2 \cos x + \sin x}{10} \right]
\end{aligned}$$

The Solution is

$$Y = Ae^x + Be^{3x} + \frac{1}{2} \left[ \frac{10 \cos 5x - 11 \sin 5x}{442} + \frac{2 \cos x + \sin x}{10} \right]$$

8. Solve  $[D^3 - 3D^2 + 4D - 2]y = e^x + \cos x$

Solution

Given  $[D^3 - 3D^2 + 4D - 2]y = e^x + \cos x$

A.E is  $m^3 - 3m^2 + 4m - 2 = 0$

$$\begin{array}{r|rrrr}
1 & 1 & -3 & 4 & -2 \\
& 0 & 1 & -2 & 2 \\
\hline
& 1 & -2 & 2 & 0
\end{array}$$

Put  $m = 1$

$$1-3+4-2=0$$

$$0=0$$

$$(m-1)(m^2-2m+2)=0$$

$$m = \frac{+2 \pm \sqrt{4-4(2)}}{2}$$

$$m = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i$$

Roots are 1,  $1 \pm i$

$$CF = Ae^x + e^x [B \cos x + C \sin x]$$

$$PI = \frac{1}{D^3 - 3D + 4D - 2} e^x + \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

$$PI = PI_1 + PI_2$$

$$PI_1 = \frac{1}{(1)^3 - 3(1)^2 + 4(1) - 2} e^x \quad \text{Replace } D \rightarrow 1$$

$$= \frac{1}{1-3+4-2} e^x \quad \text{Denominator is zero}$$

$$= \frac{x}{3D^2 - 6D + 4} e^x$$

$$= \frac{x}{3-6+4} e^x$$

$$PI_1 = xe^x$$

$$PI_2 = \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x = \frac{1}{D \cdot D^2 - 3D^2 + 4D - 2} \cos x$$

$$= \frac{1}{-D + 3 + 4D - 2} \cos x \quad \left[ \because \text{Re place } D^2 \rightarrow (-1^2) \right]$$

$$= \frac{1}{3D + 1} \cos x$$

$$= \frac{3D - 1}{(3D + 1)(3D - 1)} \cos x = \frac{3D - 1}{9D^2 - 1} \cos x$$

$$PI_2 = \frac{3 \sin x + \cos x}{10}$$

The Solution is

$$Y = Ae^x + e^x (B \cos x + C \sin x) + xe^x + \frac{1}{10} [3 \sin x + \cos x]$$

$$9. \text{ Solve } (D^2 - 4D + 4)y = e^{2x} + \cos 4x$$

### Solution

$$\text{Given } (D^2 - 4D + 4)y = e^{2x} + \cos 4x$$

$$m^2 - 4D + 4 = 0$$

$$\text{A.E is } (m-2)(m-2) = 0$$

$$m = 2, 2$$

$$\text{CF} = (Ax + B)e^{2x}$$

$$\text{PI} = \frac{1}{D^2 - 4D + 4} e^{2x} + \frac{1}{D^2 - 4D + 4} \cos 4x$$

$$\text{PI} = \text{PI}_1 + \text{PI}_2$$

$$\begin{aligned}\text{PI}_1 &= \frac{1}{D^2 - 4D + 4} e^{2x} \\ &= \frac{1}{(2)^2 - 4(2) + 4} e^{2x} \quad [\because \text{denomintor is zero}] \\ &= \frac{x}{2D - 4} e^{2x} \\ &= \frac{x}{2(2) - 4} e^{2x} = \frac{x^2}{2} e^{2x} \quad [\because \text{Denomintor is zero}]\end{aligned}$$

$$\text{PI}_1 = \frac{x^2}{2} e^{2x}$$

$$\begin{aligned}\text{PI}_2 &= \frac{1}{D^2 - 4D + 4} \cos 4x \\ &= \frac{1}{(-16) - 4D + 4} \cos 4x \quad \text{Replace } D^2 \rightarrow -16 \\ &= \frac{1}{-4D - 12} \cos 4x \\ &= \frac{-4D + 12}{16D^2 - 144} \cos 4x = \frac{-4D + 12}{16(-16) - 144} \cos 4x \\ &= \frac{-4D(\cos 4x) + 12 \cos 4x}{-256 - 144} \\ &= \frac{+16 \sin 4x + 12 \cos 4x}{-400} \\ \text{PI}_2 &= \frac{4 \sin 4x + 3 \cos 4x}{-100}\end{aligned}$$

The Solution is

$$Y = (Ax + B)e^{2x} + \frac{x^2}{2} e^{2x} - \frac{1}{100} [4 \sin 4x + 3 \cos 4x]$$

10. Solve  $(D^2 + 1)y = \cos^2 \frac{x}{2}$

Given  $(D^2 + 1)y = \cos^2 \frac{x}{2}$

A.E is  $m^2 + 1 = 0$

$m = \pm i$

CF =  $A \cos x + B \sin x$

$$\begin{aligned} \text{PI} &= \frac{1}{D^2 + 1} \cos^2 \frac{x}{2} \\ &= \frac{1}{\dots} \left[ \frac{1 + \cos x}{\dots} \right] \left[ \because \cos^2 x = \frac{1 + \cos 2x}{2} \right] \\ &= \frac{1}{2} \left[ \frac{1}{D^2 + 1} (1) + \frac{1}{D^2 + 1} \cos x \right] \\ &= \frac{1}{2} \left[ \frac{1}{D^2 + 1} e^{0x} + \frac{1}{(-1) + 1} \cos x \right] \\ &= \frac{1}{2} \left[ 1 + \frac{x}{2D} \cos x \right] \quad \because \text{Denominator is '0'} \\ &= \frac{1}{2} \left[ 1 + \frac{x}{2} \sin x \right] \quad \left[ \frac{1}{D} \cos x = \int \cos x \, dx = \sin x \right] \end{aligned}$$

$\text{PI} = \frac{1}{4} [2 + x \sin x]$

The Solution is

$Y = A \cos x + B \sin x + \frac{1}{4} [2 + x \sin x]$

### **Type – III**

$\phi(D)y = X$

Here  $X = x^m$

$P \cdot I = \frac{1}{\phi(D)} \cdot x^m$

→ Write the denominator of RHS in the form  $[1 \pm \phi_1(D)]$  by taking out the constant term or the lowest degree term of  $\phi(D)$

→ Take  $[1 \pm \phi_1(D)]$  so the numerator and apply it an  $x^m$  using Binomial expansion.



$Q(x) = x^m$ , where  $m$  is a positive integer.

$$\text{P.I} = \frac{1}{f(D)} x^m$$

Take out the lowest degree term of  $D$  in  $f(D)$  and write the other terms as  $1 + g(D)$  or  $1 - g(D)$

$$\therefore \text{P.I} = \frac{1}{D^k [1 \pm g(D)]} x^m = \frac{1}{D^k} (1 \pm g(D))^{-1} x^m$$

Since  $D^m x^m = \text{constant}$  and  $D^{m+1} x^m = 0$ , expand

$[1 \pm g(D)]^{-1}$  upto  $D^m$ , using binomial series expansion.

We have  $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$

and  $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots$

$$(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

and  $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$

**Solve  $(D^3 - D^2 - D + 1)y = 1 + x^2$**

**Solution**

The given equation is

$$(D^3 - D^2 - D + 1)y = 1 + x^2$$

To find the complementary function, solve  $(D^3 - D^2 - D + 1)y = 0$

Auxiliary equation is  $m^3 - m^2 - m + 1 = 0$

$$\Rightarrow m^2(m - 1) - (m - 1) = 0$$

$$\Rightarrow (m - 1)(m^2 - 1) = 0$$

$$\Rightarrow (m - 1)(m - 1)(m + 1) = 0 \Rightarrow m = +1, +1, -1$$

The roots are real with two roots equal and the third different.

$$\therefore \text{C.F} = C_1 e^{-x} + (C_2 + C_3 x) e^x$$

$$\begin{aligned}
\text{P.I} &= \frac{1}{D^3 - D^2 - D + 1}(1 + x^2) \\
&= \frac{1}{1 - (D + D^2 - D^3)}(1 + x^2) \\
&= [1 - (D + D^2 - D^3)]^{-1}(1 + x^2) \\
&= [1 + (D + D^2 - D^3) + (D + D^2 - D^3)^2 + \dots](1 + x^2) \\
&= (1 + D + 2D^2)(1 + x^2) \quad [\text{Taking terms upto } D^2] \\
&= 1 + x^2 + D(1 + x^2) + 2D^2(1 + x^2) \\
&= 1 + x^2 + 2x + 2 \cdot 2 = x^2 + 2x + 5
\end{aligned}$$

$\therefore$  the general solution is  $y = \text{C.F} + \text{P.I}$

$$y = C_1 e^{-x} + (C_2 + C_3 x)e^x + x^2 + 2x + 5$$

**Solve  $(D^2 + 4)y = x^4 + \cos^2 x$**

**Solution**

The given equation is

$$\begin{aligned}
(D^2 + 4)y &= x^4 + \cos^2 x \\
&= x^4 + \frac{1 + \cos 2x}{2} = \left(x^4 + \frac{1}{2}\right) + \frac{\cos 2x}{2}
\end{aligned}$$

To find the complementary function, solve  $(D^2 + 4)y = 0$

Auxiliary equation is  $m^2 + 4 = 0$

$$\Rightarrow m^2 = -4 \Rightarrow m = \pm 2i$$

The roots are complex with  $\alpha = 0$  and  $\beta = 2$

$$\therefore \text{C.F} = C_1 \cos 2x + C_2 \sin 2x$$

$$\begin{aligned}
\text{P.I}_1 &= \frac{1}{D^2 + 4} \left(x^4 + \frac{1}{2}\right) \\
&= \frac{1}{4} \left(1 + \frac{D^2}{4}\right)^{-1} \left(x^4 + \frac{1}{2}\right) \\
&= \frac{1}{4} \left[1 - \frac{D^2}{4} + \frac{D^4}{16}\right] \left[x^4 + \frac{1}{2}\right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \left[ x^4 + \frac{1}{2} - \frac{1}{4} D^2 \left( x^4 + \frac{1}{2} \right) + \frac{1}{16} D^4 \left( x^4 + \frac{1}{2} \right) \right] \\
&= \frac{1}{4} \left[ x^4 + \frac{1}{2} - \frac{4 \cdot 3}{4} x^2 + \frac{1}{16} 4 \cdot 3 \cdot 2 \cdot 1 \right] \\
&= \frac{1}{4} \left[ x^4 - 3x^2 + \frac{1}{2} + \frac{3}{2} \right] \\
&= \frac{1}{4} [x^4 - 3x^2 + 2]
\end{aligned}$$

$$\begin{aligned}
\text{P.I}_2 &= \frac{1}{2} \cdot \frac{1}{D^2 + 4} \cos 2x \\
&= \frac{1}{2} \cdot \frac{x}{2} \int \cos 2x \, dx && [\because f(-2)^2 = 0] \\
&= \frac{x}{4} \left[ \frac{\sin 2x}{2} \right] = \frac{x \sin 2x}{8}
\end{aligned}$$

$\therefore$  the general solution is

$$\begin{aligned}
y &= \text{C.F} + \text{P.I}_1 + \text{P.I}_2 \\
&= C_1 \cos 2x + C_2 \sin 2x + \frac{1}{4}(x^4 - 3x^2 + 2) + \frac{x \sin 2x}{8}
\end{aligned}$$

**Solve**  $(D^2 + D)y = x^2 + 2x + 4$

**Solution**

The given equation is

$$(D^2 + D)y = x^2 + 2x + 4$$

To find the complementary function, solve  $(D^2 + D)y = 0$

Auxiliary equation is  $m^2 + m = 0$

$$\Rightarrow m(m + 1) = 0 \quad \therefore m = 0, -1$$

The roots are real and different.

$$\therefore \text{C.F} = C_1 e^{0x} + C_2 e^{-x} = C_1 + C_2 e^{-x}$$

$$\begin{aligned} \text{P.I} &= \frac{1}{D^2 + D}(x^2 + 2x + 4) \\ &= \frac{1}{D} \frac{1}{(1 + D)}(x^2 + 2x + 4) \\ &= \frac{1}{D}(1 + D)^{-1}(x^2 + 2x + 4) \\ &= \frac{1}{D}(1 - D + D^2)(x^2 + 2x + 4) \\ &= \frac{1}{D}[(x^2 + 2x + 4) - D(x^2 + 2x + 4) + D^2(x^2 + 2x + 4)] \\ &= \frac{1}{D}[x^2 + 2x + 4 - (2x + 2) + 2] \\ &= \frac{1}{D}[x^2 + 4] = \int (x^2 + 4) dx = \frac{x^3}{3} + 4x \end{aligned}$$

$\therefore$  the general solution is  $y = \text{C.F} + \text{P.I}$

$$= C_1 + C_2 e^{-x} + \frac{x^3}{3} + 4x$$

### **Discussion Questions**

Try to answer the following questions

1. Why we use second order linear ordinary differential equations?
2. Why do you solve second order linear ordinary differential equations?